

## Chatbot MQL Automation | Troubleshooting & Update Report | 6.25.26

**Project:** Atlas HXM HubSpot Customer Agent / Maia MQL Automation

**Workflow:** Chatbot MQL Automation - Final

**System:** HubSpot + n8n + Azure OpenAI

**Status:** Built and tested through full closed-chat enrichment flow

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### 1. Executive Summary

The chatbot MQL automation was updated to process closed HubSpot Customer Agent conversations, retrieve the associated contact and conversation thread data, pull the full message transcript, analyze the transcript with AI, classify the contact as MQL or Non-MQL, and update the HubSpot contact record with structured chatbot intelligence.

The workflow now supports transcript capture, buying-need interpretation, country extraction, hiring timeline extraction, workforce setup analysis, EOR intent, MQL classification, AI-generated email body output, session ID storage, and conditional lifecycle-stage updates for MQLs.

A major architecture improvement was the addition of a dedicated preparation layer between the AI node and HubSpot update node. This prevents raw AI output from being sent directly to HubSpot and allows invalid or blank values to be cleaned before the contact update runs.

### 2. Final Workflow Architecture

The final intended flow is:

- HubSpot Closed Conversation Webhook
- Find Contact
- Get Conversation Thread
- Get Thread Messages
- Normalize Conversation + Transcript
- Analyze + Draft Email
- Prepare HubSpot Contact Fields
- Create or Update Contact

### Node Purpose Summary

| Node                                       | Purpose                                                                                                |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>HubSpot Closed Conversation Webhook</b> | Receives the closed chatbot conversation payload from HubSpot, including contact ID and thread ID.     |
| <b>Find Contact</b>                        | Searches HubSpot CRM for the contact using hs_object_id.                                               |
| <b>Get Conversation Thread</b>             | Retrieves HubSpot conversation thread metadata. Used for validation and thread/session context.        |
| <b>Get Thread Messages</b>                 | Pulls the conversation messages from the HubSpot Conversations API.                                    |
| <b>Normalize Conversation + Transcript</b> | Sorts messages, builds a clean transcript, extracts email, phone, company, domain, and session fields. |
| <b>Analyze + Draft Email</b>               | Uses Azure OpenAI to analyze the transcript and return structured MQL fields.                          |
| <b>Structured Output Parser</b>            | Attempts to enforce consistent JSON from the AI node.                                                  |
| <b>Prepare HubSpot Contact Fields</b>      | Cleans, flattens, truncates, and normalizes AI output before HubSpot update.                           |
| <b>Create or Update Contact</b>            | Updates the HubSpot contact using mapped standard and custom properties.                               |

### 3. Key Changes Made

#### A. Fixed Contact Lookup

The workflow now looks up contacts by HubSpot record ID using:

hs\_object\_id

This fixed the earlier issue where broad contact searches returned unrelated records.

The contact search now uses the CRM search endpoint:

POST <https://api.hubapi.com/crm/v3/objects/contacts/search>

with a filter on:

propertyName: hs\_object\_id

operator: EQ

value: webhook body hs\_object\_id

## **B. Fixed Thread and Message Retrieval**

The workflow was updated to use the HubSpot Conversations API to retrieve the thread and messages using the thread ID from the webhook.

Thread endpoint:

[https://api.hubapi.com/conversations/v3/conversations/threads/{hs\\_thread\\_id}](https://api.hubapi.com/conversations/v3/conversations/threads/{hs_thread_id})

Messages endpoint:

[https://api.hubapi.com/conversations/v3/conversations/threads/{hs\\_thread\\_id}/messages](https://api.hubapi.com/conversations/v3/conversations/threads/{hs_thread_id}/messages)

This resolved the earlier issue where only conversation metadata was returned and no transcript was available. The transcript must come from the /messages endpoint, not the base thread endpoint.

## **C. Added Transcript Normalization**

A new Normalize Conversation + Transcript code node was added.

This node:

- Sorts messages oldest-to-newest.
- Labels messages as Visitor or Atlas HXM assistant.
- Removes HTML and extra whitespace.
- Builds a clean transcript.
- Extracts visitor messages separately.
- Extracts email, phone, company name, company domain, website, and contact details when available.

- Creates clean session fields such as chatbot\_session\_id and chatbot\_last\_session\_date.

This prevents the AI from analyzing raw HubSpot message payloads with unnecessary metadata.

#### **D. Added Structured AI Analysis**

The AI prompt was expanded to return structured JSON for all required chatbot fields.

The AI now analyzes:

- MQL status
- Confidence
- Buying need
- Pain points
- Hiring timeline
- Countries of interest
- EOR intent
- Workforce setup
- Visitor questions asked
- Recommended next step
- Disqualification reason
- Follow-up email body
- MQL lifecycle output
- Number of employees hiring
- Current EOR usage
- Current EOR challenge information

#### **E. Updated MQL Classification Rules**

The MQL logic was tightened.

A contact should only be classified as MQL when there is a work email or clear company identity plus B2B intent around:

- Global hiring
- EOR
- Payroll
- Compliance
- Benefits
- Pricing

- Demo request
- Country coverage
- Entity avoidance or setup
- Hiring timeline
- Employee headcount
- International workforce support

The AI was also instructed not to classify contacts as MQL based only on Maia's chatbot responses. Only visitor messages count as buying intent.

#### **F. Added AI Email Body Output**

The workflow now outputs the AI-generated email body to:

ai\_mql\_response\_1

The email output rules were refined so the AI returns only the middle body content.

The email body should not include:

- Greeting
- Contact name
- Intro
- Subject line
- Signature
- Sign-off

This allows the text to be inserted into an existing formatted email template.

#### **G. Added Conditional Lifecycle Stage Logic**

The workflow now creates a lifecycle-stage output only when:

chatbot\_mql\_status = MQL

When MQL, the output is:

lifecyclestage = marketingqualifiedlead

For Non-MQL contacts, lifecycle stage remains blank and should not be sent to HubSpot. This prevents Non-MQL records from overwriting existing lifecycle values.

#### **H. Added Field Preparation Layer**

A Prepare HubSpot Contact Fields node was added between the AI node and the HubSpot update node.

This node:

- Parses AI output.
- Handles JSON parsing failures.
- Merges normalized contact/session data with AI output.
- Truncates long values.
- Joins country arrays into HubSpot-safe strings.
- Deletes blank values before HubSpot update.
- Prevents empty values from overwriting existing CRM data.
- Adds MQL-only fields when applicable.

This was a major stability improvement.

#### 4. HubSpot Properties Added or Updated

The workflow now supports updating the following HubSpot properties.

##### Core Chatbot Fields

| Property                | Purpose                                                   |
|-------------------------|-----------------------------------------------------------|
| chatbot_transcript      | Full cleaned conversation transcript.                     |
| chatbot_wses            | Workforce setup / workforce scenario.                     |
| chatbot_timeline        | Hiring, EOR, payroll, implementation, or buying timeline. |
| chatbot_countries       | Countries mentioned or implied by the visitor.            |
| chatbot_eor_intent      | Whether the visitor showed EOR intent.                    |
| chatbot_mql_status      | MQL or Non-MQL classification.                            |
| chatbot_session_id      | HubSpot conversation thread ID.                           |
| chatbot_buyer_need      | Interpreted buying need from the transcript.              |
| chatbot_pain_points     | Pain points, blockers, or risks from the transcript.      |
| ai_chatbot_processed    | Indicates the AI analysis was processed.                  |
| chatbot_questions_asked | Visitor questions only, not Maia's questions.             |

|                                 |                                           |
|---------------------------------|-------------------------------------------|
| chatbot_last_session_date       | Last chatbot session/message date.        |
| chatbot_recommended_next_step   | Recommended next sales/system action.     |
| chatbot_disqualification_reason | Reason for Non-MQL status, if applicable. |

### MQL and Sales Fields

| Property                                                                 | Purpose                                                    |
|--------------------------------------------------------------------------|------------------------------------------------------------|
| ai_mql_response_1                                                        | AI-generated follow-up email body.                         |
| lifecyclestage                                                           | Set to marketingqualified only for MQLs.                   |
| hs_lead_status                                                           | Set to NEW for MQLs.                                       |
| customer_agent_mql_date                                                  | Date the chatbot classified contact as MQL.                |
| mql__number_of_employees_hiring                                          | Number of employees the visitor wants to hire/pay/support. |
| are_you_currently_using_an_employer_of_record_eor_service_               | Whether visitor currently uses an EOR service.             |
| mql__what_challenges_you_re_facing_with_your_current_eor_service__multi_ | Current EOR provider challenges, if applicable.            |

### Standard Contact Fields

| Property  | Purpose             |
|-----------|---------------------|
| email     | Contact email.      |
| firstname | Contact first name. |
| lastname  | Contact last name.  |

|          |                                                         |
|----------|---------------------------------------------------------|
| phone    | Phone number from CRM or transcript.                    |
| company  | Company name from CRM, transcript, or domain inference. |
| domain   | Company domain.                                         |
| website  | Website URL.                                            |
| jobtitle | Job title from CRM or transcript.                       |

## 5. Issues Encountered and Fixes

### Issue 1: Wrong Contact Returned

**Problem:** HubSpot returned an unrelated contact because the contact search was too broad.

**Fix:** Search was updated to filter by:

hs\_object\_id

using the contact ID from the webhook.

### Issue 2: Thread Endpoint Returned Metadata Only

**Problem:** The base thread endpoint returned metadata like ID, status, created date, and associated contact ID, but not the transcript.

**Fix:** Added the messages endpoint:

/conversations/v3/conversations/threads/{threadId}/messages

This endpoint provides the actual chat messages needed to build the transcript.

### Issue 3: AI Output Did Not Fit Required Format

**Problem:** The AI sometimes returned malformed JSON or schema-like labels instead of valid values.

**Fixes:**

- Added a Structured Output Parser.

- Tightened the AI prompt to require valid JSON only.
- Added explicit output requirements.
- Added parser compatibility rules.
- Added a Prepare HubSpot Contact Fields node to parse and clean output before HubSpot update.

Recommended fallback:

- Set the AI node's error behavior to continue where possible.
- Let the Prepare node create safe default values if the AI output fails.

#### **Issue 4: HubSpot Property Values Were Not Valid**

**Problem:** HubSpot rejected updates because some values did not match property types or allowed dropdown values.

Most likely causes included:

- Date fields receiving full ISO datetime instead of date-only values.
- Dropdowns receiving lowercase yes/no/unknown when HubSpot expected exact option values.
- Multi-select fields receiving free text instead of allowed option values.
- Empty or invalid optional fields being sent.

**Fixes:**

- Delete blank values before HubSpot update.
- Do not send optional fields if unavailable.
- Do not send free text into multi-select properties.
- Only populate current EOR challenge fields when the visitor actually uses a current EOR and mentions a challenge.
- Use date-only formatting when the HubSpot property is a date picker.

#### **Issue 5: Questions Asked Field Included Maia's Questions**

**Problem:** The AI initially placed Maia's qualifying questions into chatbot\_questions\_asked.

**Fix:** Prompt was updated to specify that chatbot\_questions\_asked must include only questions asked by the visitor. If the visitor did not ask a question, the field should be empty.

## Issue 6: Chatbot Timeline Was Misinterpreted

**Problem:** chatbot\_timeline was initially treated as a conversation timeline/log.

**Fix:** The field was redefined to mean the buyer's hiring, payroll, EOR, implementation, or purchase timeline.

Example:

65 days

not:

Visitor asked about EOR, then Maia asked about country, then visitor gave timeline.

## Issue 7: Follow-Up Email Included Intro/Signature

**Problem:** The generated email body initially included greetings or intro-style copy.

**Fix:** Prompt was updated so follow\_up\_email\_body and ai\_mql\_response\_1 return only middle-body copy with no greeting, no intro, no subject, no signature, and no sign-off.

## 6. Final Field Logic

### MQL Logic

If:

chatbot\_mql\_status = MQL

then set:

lifecyclestage = marketingqualifiedlead

hs\_lead\_status = NEW

customer\_agent\_mql\_date = today

If:

chatbot\_mql\_status = Non-MQL

then do not send lifecyclestage.

### AI Email Logic

If MQL and there is a clear buyer need:

ai\_mql\_response\_1 = follow\_up\_email\_body

If Non-MQL or buyer need is unclear:

ai\_mql\_response\_1 = blank

### **EOR Challenge Logic**

If visitor is currently using an EOR and mentions challenges:

mql\_\_\_what\_challenges\_you\_re\_facing\_with\_your\_current\_eor\_service\_\_\_multi\_ = valid  
HubSpot option value

If visitor is starting fresh or not using an EOR:

mql\_\_\_what\_challenges\_you\_re\_facing\_with\_your\_current\_eor\_service\_\_\_multi\_ = blank

### **Optional Field Logic**

If a field is empty, missing, null, or invalid:

Do not send it to HubSpot.

This prevents update failures and avoids overwriting existing CRM data.

## **7. Current Known Risk Areas**

### **A. Dropdown and Multi-Select Option Values**

HubSpot dropdown, radio, and multi-checkbox properties require exact internal option values.

Fields to verify:

chatbot\_eor\_intent

chatbot\_mql\_status

ai\_chatbot\_processed

chatbot\_lead\_temperature

are\_you\_currently\_using\_an\_employer\_of\_record\_\_\_eor\_\_\_service\_

mql\_\_\_what\_challenges\_you\_re\_facing\_with\_your\_current\_eor\_service\_\_\_multi\_

### **B. Date Property Formatting**

If a HubSpot property is a date picker, send:

YYYY-MM-DD

not full ISO datetime.

Example:

2026-06-02

### **C. Native HubSpot Node Behavior With Blank Values**

The native HubSpot Create or Update Contact node may still attempt to send mapped blank fields. For production, optional fields should either be omitted before mapping or split into separate update nodes.

Recommended production structure:

Prepare HubSpot Contact Fields

→ HubSpot Base Update

→ IF isMQL

→ HubSpot MQL Update

## **8. Recommended Production Architecture**

### **Base Update Node**

Always update safe fields:

email

firstname

lastname

phone

company

domain

website

jobtitle

chatbot\_transcript

chatbot\_session\_id

chatbot\_last\_session\_date

ai\_chatbot\_processed

chatbot\_mql\_status

chatbot\_buyer\_need

chatbot\_recommended\_next\_step

maia\_chat\_summary

### **MQL-Only Update Node**

Only run when:

isMQL = true

Update:

lifecyclestage

hs\_lead\_status

customer\_agent\_mql\_date

mql\_\_number\_of\_employees\_hiring

are\_you\_currently\_using\_an\_employer\_of\_record\_\_eor\_\_service\_\_

ai\_mql\_response\_1

### **Conditional / Optional Fields**

Only update if valid:

mql\_\_what\_challenges\_you\_re\_facing\_with\_your\_current\_eor\_service\_\_multi\_\_

chatbot\_pain\_points

chatbot\_questions\_asked

chatbot\_disqualification\_reason

## **9. Troubleshooting Checklist**

### **If HubSpot update fails with “Property values were not valid”**

Check:

1. Is a dropdown receiving a value that does not match HubSpot internal option values?
2. Is a date field receiving full ISO datetime instead of YYYY-MM-DD?
3. Is a multi-select property receiving free text?
4. Are duplicate mappings sending different values to the same property?
5. Is a blank mapped field overwriting a required property?
6. Is lifecyclestage being sent blank for Non-MQL records?

### **If AI node fails with “Model output does not fit required format”**

Check:

1. Is the Structured Output Parser schema using real example values, not labels like MQL or Non-MQL?
2. Is the AI returning markdown or code fences?
3. Is the AI returning an array instead of a single object?
4. Is a field returning the wrong type, such as string instead of array?

5. Is the model temperature too high for structured output?

Recommended fix:

Set temperature to 0 and allow the Prepare node to handle fallback parsing.

### **If transcript is wrong or backwards**

Check:

1. Are messages sorted oldest-to-newest?
2. Is the transcript using the /messages endpoint?
3. Are sender labels correctly identifying Visitor vs Atlas HXM assistant?
4. Is the thread ID coming from the correct webhook execution item?

### **If MQL classification is wrong**

Check:

1. Did the visitor provide actual buying intent?
2. Is the AI counting Maia's messages as buyer intent?
3. Did the visitor say "no" to global hiring or EOR needs?
4. Is confidence below 0.65?
5. Is the visitor a job seeker, support request, vendor, partner, or spam?

### **If company, phone, or domain are missing**

Check:

1. Did Find Contact request those properties?
2. Did the visitor provide them in the transcript?
3. Did Normalize extract them?
4. Are the HubSpot node mappings using the prepared top-level fields, not raw AI fields?

## **10. Final Outcome**

The workflow now provides a complete closed-chat MQL automation for Atlas HXM.

It can:

- Receive closed HubSpot Customer Agent chat data.
- Identify the associated contact.
- Pull the conversation thread and messages.
- Build a clean transcript.
- Analyze the transcript with AI.

- Determine MQL vs Non-MQL.
- Interpret buyer need, timeline, country, EOR intent, workforce setup, and employee count.
- Generate a reusable AI email body.
- Update the HubSpot contact with structured fields.
- Set lifecycle stage to marketingqualifiedlead when qualified.
- Avoid blocking updates when optional fields are empty.

This creates a scalable foundation for routing qualified chatbot conversations into sales follow-up and round-robin assignment workflows.